

Master's Thesis - Research Outline

1. Motivation

- An empirical review of all mergers challenged by the U.S. FTC between 1995 and 2008 finds that innovation concerns played a role in $\frac{1}{3}$ of the challenged cases ([Kern, Dewenter and Kerber, 2016](#))
- The pharmaceutical sector is subject to heightened antitrust scrutiny due to **structural factors** (high concentration, repeated consolidation) and **market characteristics** (exceptionally high R&D-to-revenue ratios and frequent M&A activity)
- In 2023 the [FTC](#) established a *Multilateral Pharmaceutical Merger Task Force* to reassess how pharmaceutical mergers are evaluated. One central question guiding this effort is: “What is the full range of a pharmaceutical merger’s effects on innovation?”

1.1 Previous Empirical Results

- Prior **firm-level** studies document substantial post-merger declines in pharmaceutical R&D output. *Ornaghi (2009)* finds that research output falls by **10–20%** in the three years following an acquisition, while *Haucap et al. (2019)*, using a more comprehensive dataset, report a **30%** reduction in patenting activity
- **Inventor-level** evidence shows substantial post-merger talent loss and productivity decline: Verginer et al. find inventors are **20% more likely** to leave after an acquisition, while Cassi & Ornaghi (2025) show that acquisitions **raise exit rates by 7.6–17.5 p.p.** and significantly reduce patent output: 2.5 foregone patents per exiting inventor ($\approx 38\%$ loss), 2 fewer patents for movers ($\approx 30\%$ productivity drop), and ~ 1 patent fewer for stayers

2. Research Question

What is the **impact of an acquisition** on the post-merger patenting quantity, citation-weighted quality, and *field-switching behavior (more explorative)* of target-firm inventors who remain in the merged organization, compared to similar inventors not exposed to the merger?

2.1 Potential reasons for lower patent output post-merger

- **Weaker support for internal ventures:** After a merger, firms often shift focus toward existing product lines rather than funding early-stage internal ideas, reducing the “championing” of new projects (Hitt, Hoskisson, Ireland & Harrison, 1991)
- **Integration frictions:** The acquirer’s organizational structure can disrupt established routines in the target firm, forcing researchers to spend time navigating new processes (*bureaucracy*) rather than discovering new ideas (Kapoor & Lim, 2007)

- **Loss of status (“Conqueror’s Effect”):** Inventors that were central or influential in the pre-merger R&D setup may lose their status in the post-merger, which lowers motivation and thus reduces their research output (Paruchuri, Nerkar & Hambrick, 2006)
- **Reduced bargaining power (*out of scope, but interesting*):** In highly specialized sectors like pharma, consolidation lowers external job options for researchers, weakening their bargaining power and dampening incentives (He & Xue, 2025)
- **Disrupted knowledge networks:** Integration breaks informal communication channels, limiting researchers’ access to the *tacit knowledge* needed for efficient problem-solving and slowing innovation among those who stay (Ernst & Vitt, 2000)

3. Data [Patents + M&A] (*these are your data sources*)

- **Patent Data:** European Patent Office Data for inventors in the pharmaceutical industry working there between 1978 and 2015
- **M&A Data:** Orbis and Zephyr, published by Bureau van Dijk

4. Identification Strategy (*rough idea - exploratory*)

4.1 Target Parameter

Target parameter: **ATT on post-acquisition patent output** for inventors coming from the target firm (*inventors from the acquirer do not face the same degree of org. change*)

- Treated Group: inventors employed by target firms in the 3–5 years before an acquisition
- Control Group: inventors in non-acquired firms with similar characteristics (*IPC technological profile (cosine similarity), age, patenting behavior pre-acquisition*)

4.2 Empirical Strategy [*exploratory*]

1. **Inventor Identification:** Disambiguate names and assign each inventor a unique ID to track their full career and link all patents listing them as authors. (*1st idea for Affiliation handling*)
2. **Firm Group Identification:** Create a group-level code linking all subsidiaries to the parent company
3. **Status Definition:** Classify inventors post-merger as $D = 1$ (stayer) if they continue patenting for the acquiring group at least until $t_{acq} + x$, and $D = 0$ (leaver) otherwise [*for this research q , we don’t need to differentiate between exit and leave*]
4. **Matching (Two-Stage):** First match treated and untreated firms based on organizational and technological characteristics (one-to-many). Then match treated inventors 1-to-1

with comparable inventors drawn from the matched firm pool

5. **Estimation:** Among stayers, compare their patent output, citation-weighted quality, and field switching over the 3- and 5-year post-merger windows to estimate how the acquisition affects the innovation performance of those who remain [(modern) DID?]

P5 is fully exploratory at the moment and I will think more of this when I have data